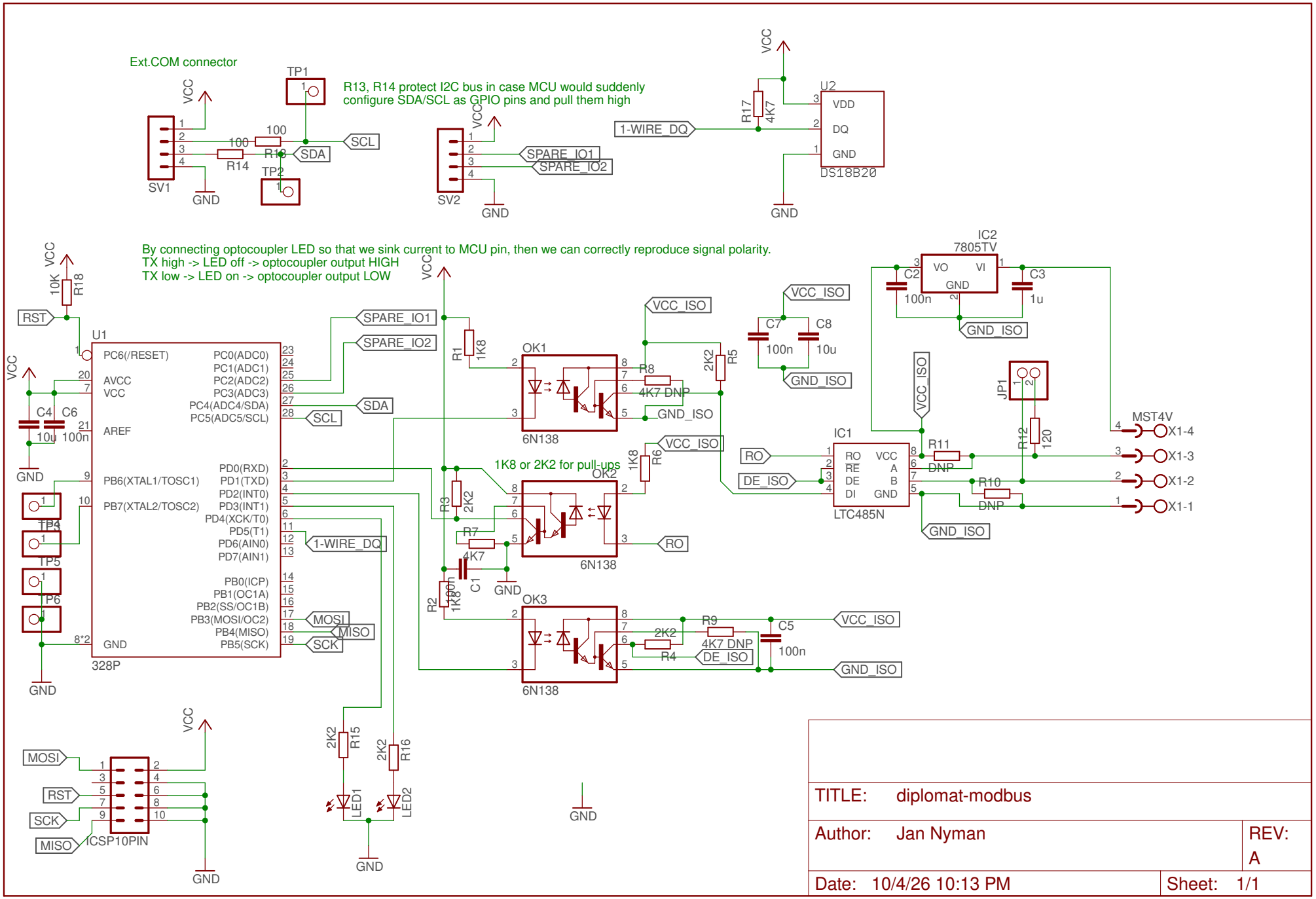
Ext.COM connector
TP1
VCC
SV1
100
R13
R14
SCL
SDA
TP2
GND
R13, R14 protect I2C bus in case MCU would suddenly
configure SDA/SCL as GPIO pins and pull them high
SV2
SPARE_IO1
SPARE_IO2
1-WIRE_DQ
R17
4K7
U2
VDD
DQ
GND
DS18B20
By connecting optocoupler LED so that we sink current to MCU pin, then we can correctly reproduce signal polarity.
TX high -> LED off -> optocoupler output HIGH
TX low -> LED on -> optocoupler output LOW
10K
R18
RST
U1
PC6(/RESET)
PC0(ADC0)
PC1(ADC1)
PC2(ADC2)
PC3(ADC3)
PC4(ADC4/SDA)
PC5(ADC5/SCL)
AVCC
VCC
AREF
C4
C6
10u
100n
PB6(XTAL1/TOSC1)
PB7(XTAL2/TOSC2)
PD0(RXD)
PD1(TXD)
PD2(INT0)
PD3(INT1)
PD4(XCK/T0)
PD5(T1)
PD6(AIN0)
PD7(AIN1)
PB0(ICP)
PB1(OC1A)
PB2(SS/OC1B)
PB3(MOSI/OC2)
PB4(MISO)
PB5(SCK)
8*2
328P
TP4
TP5
TP6
1-WIRE_DQ
MOSI
MISO
SCK
R1
1K8
OK1
6N138
R8
4K7 DNP
2K2
R5
VCC_ISO
GND_ISO
1K8 or 2K2 for pull-ups
R3
R7
4K7
C1
100n
R2
OK2
OK3
R6
R4
R9
C5
DE_ISO
RO
IC2
7805TV
VO
VI
C2
C3
1u
C7
C8
10u
IC1
RO
RE
DE
DI
VCC
A
B
GND
LTC485N
R11
DNP
R10
JP1
R12
120
MST4V
X1-4
X1-3
X1-2
X1-1
ICSP10PIN
R15
R16
LED1
LED2
TITLE: diplomat-modbus
Author: Jan Nyman
REV: A
Date: 10/4/26 10:13 PM
Sheet: 1/1